

- 18** A vertical tube is completely filled with water. A small sound source of constant frequency is held a little above the open upper end and water is run out from the lower end.

A number of resonance positions are detected. The second position was detected when the water surface is 12 cm below the top of the tube and another occurs at 36 cm.

At which of the following distances should resonance also be detected?

- A** 24 cm **B** 26 cm **C** 44 cm **D** 48 cm

Ans: C

The wavelength can be deduced to be 16 cm as 12 cm below the water surface is the 2nd occurrence of resonance. 36 cm below water surface is the 5th occurrence. Other possible resonance positions include 4 cm, 20 cm, 28 cm, 44 cm, 52 cm below the water surface.

